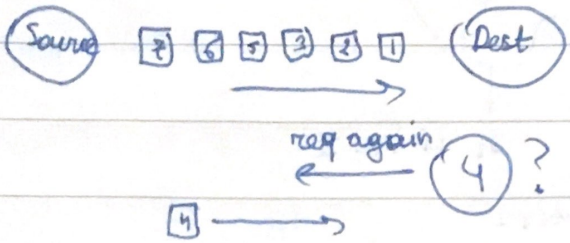
TCP



port (0-65535)

well-known - (0-1033)

registered - (1034-49151)

dynamic/private (49152-65535)

used by computer

UDP

connection-less

src → unacknowledged → dest

ICMP - diagnostics

ping

traceroute

IP - connectionless service

DNS - udp: 53 - client-server

tcp: 53 - inter-server

SMTP - tcp: 25

POPv3 - tcp: 110

recieving email

IMAP - tcp: 143

manage email on server

Telnet - cleartext

tcp: 23

SSH - tcp: 22 → tunnel + FTP = SFTP

FTP - tcp: 21 (commands)

tcp: 20 (data transfer) ↑ data over commands

cleartext

TFTP - udp: 69 data transfer

trivial

Отноо

DHCP - IP address assignment
Dynamic host config
67
68

HTTP : 80

HTTP + TLS = HTTPS : 443

RDP - remote desktop tcp: 3389

LDAP - lightweight directory access tcp: 389

LDAPS - tcp: 636

NTP port: 123

VOIP:
SIP - session initiation [5060, 5061]
H323 → 1720

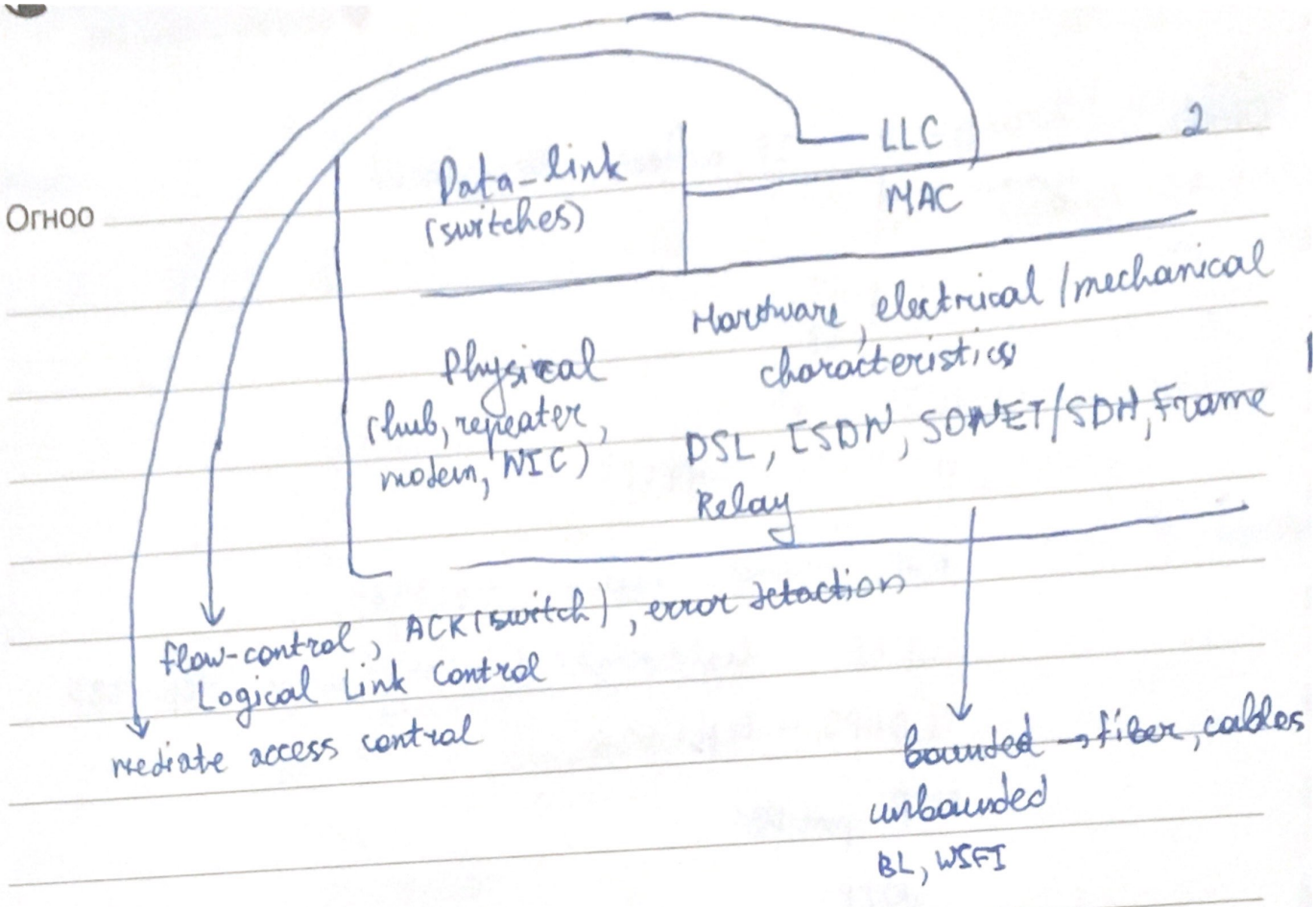
SMB - server message block ports: [445]

OSI Model

application functioning block

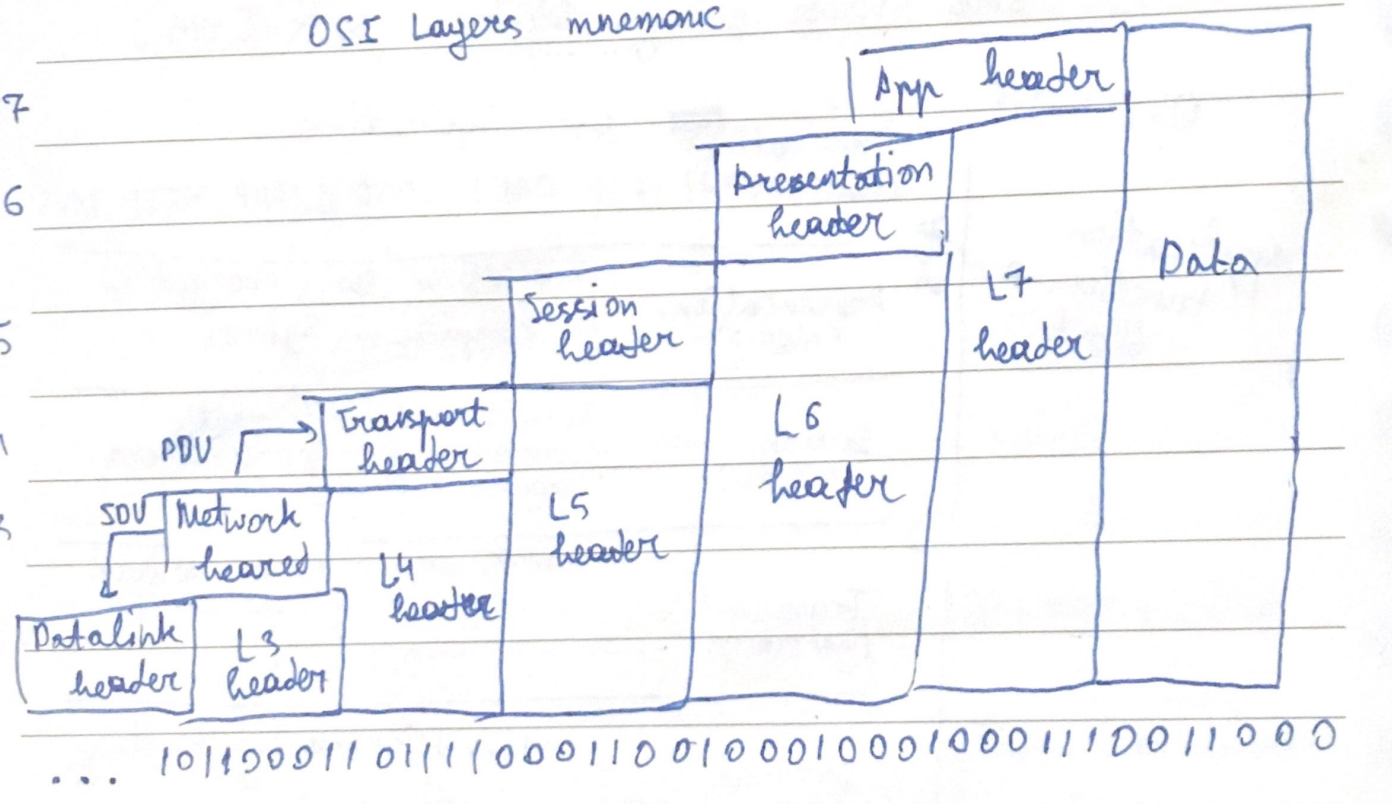
Application (hosts, NGFW)	User applications FTP, POP3, SMTP, IMAP, HTTP, DNS		7
Presentation (NGFW)	Formatting, Data encryption compression, Syntax	Data	6
Session (NGFW)	Connection establishment communication parameters maintaining		5
Transport (gateways)	move data along the network segmenting, sequencing TCP, UDP		4
Network (router, L3 switch)	Packet forwarding, routing, addressing IP, ICMP, OSPF, RIPv2, BGP		3

Огноо



All people seem to need data processing.

OSI Layers mnemonic



Broadcast

one-to-all communication

Unicast

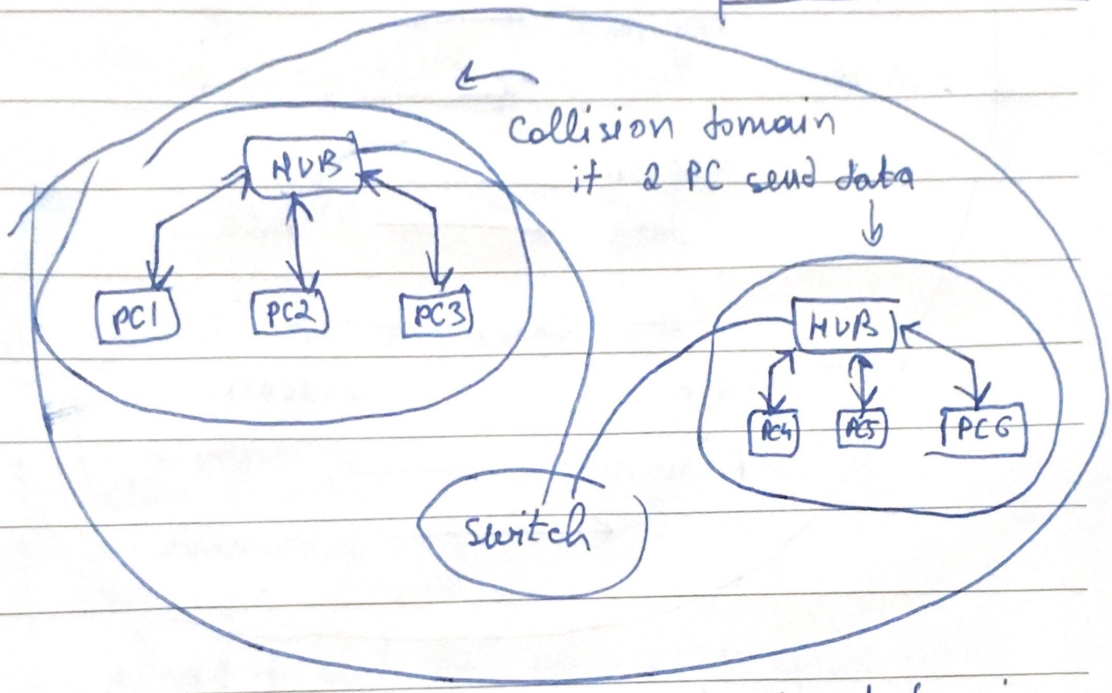
one-to-one communication

Multicast

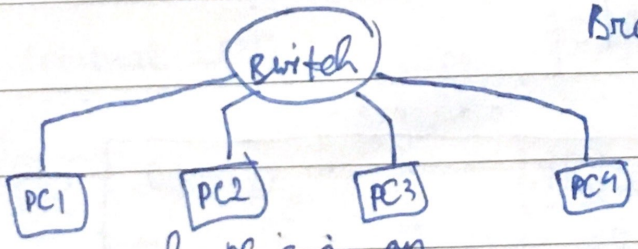
one-to-many (specific group)

Anycast

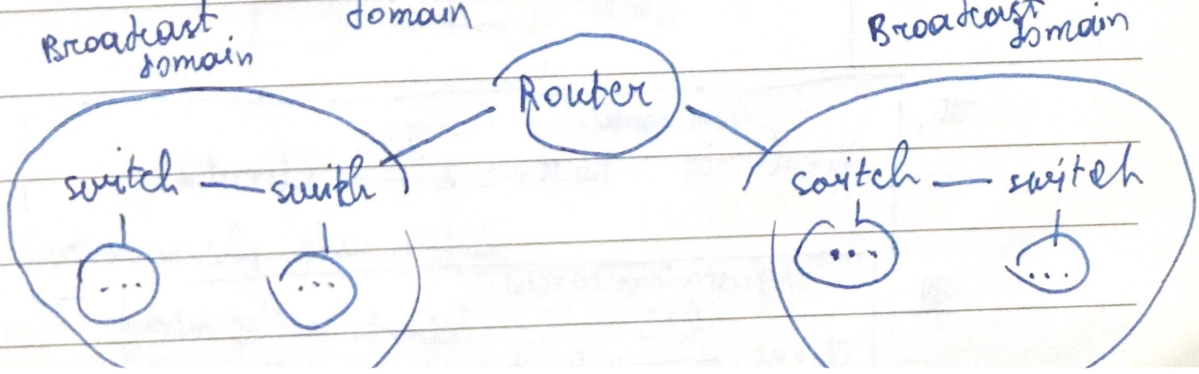
one-to-the-best communication



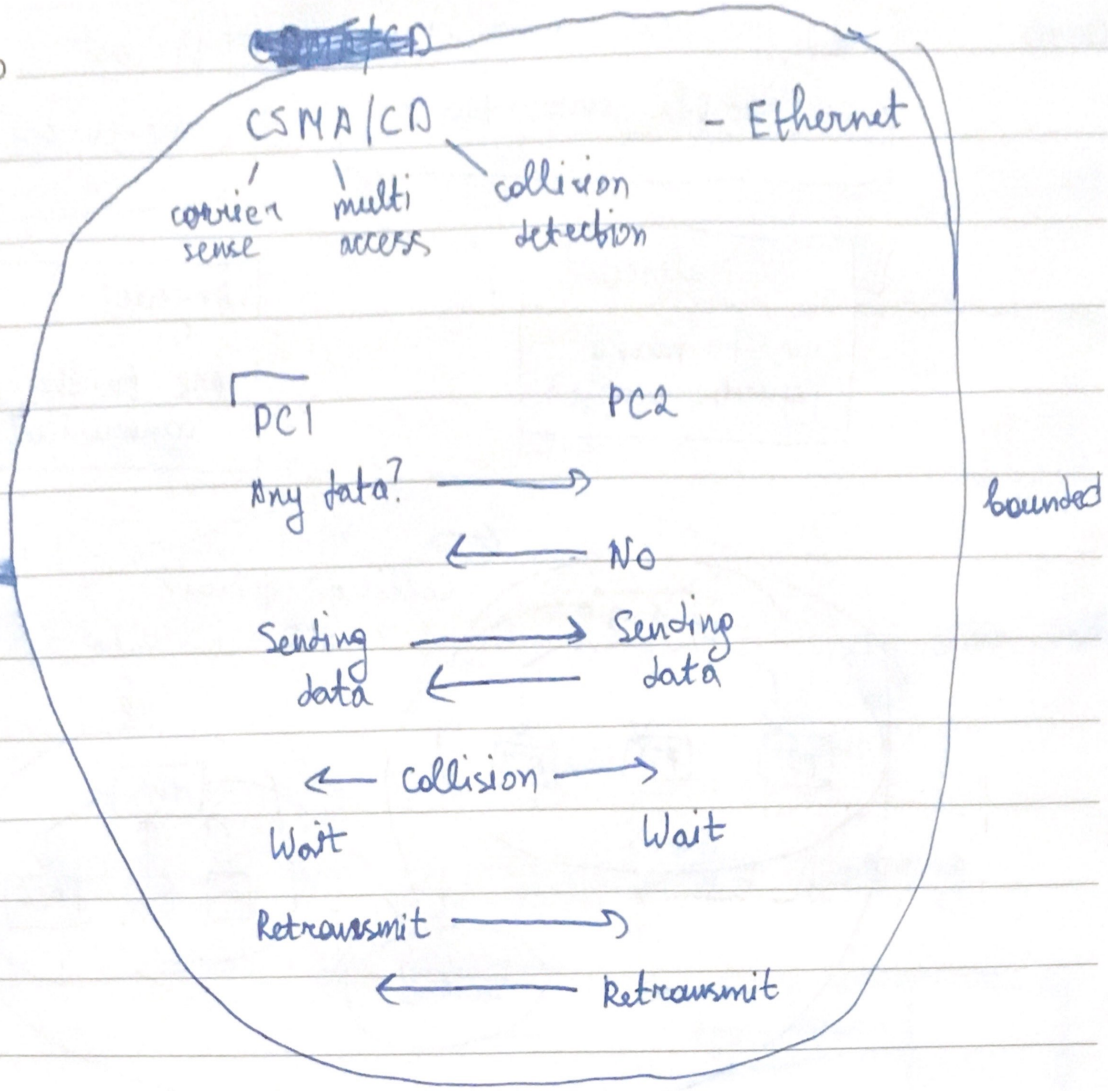
Broadcast domain



each PC is in an isolated collision domain

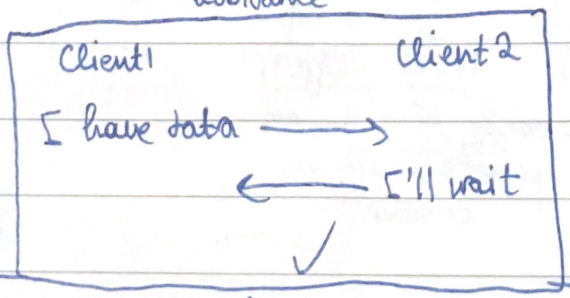


Огноо

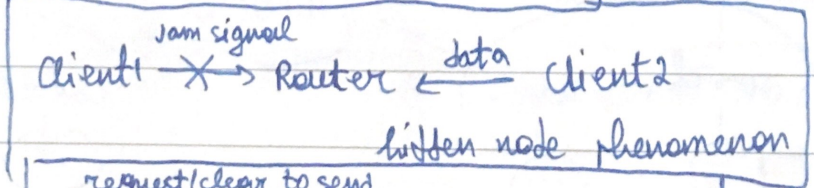


CSMA/CA
↳ collision avoidance

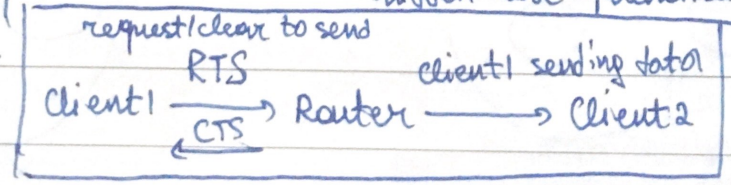
I.



II.



III.

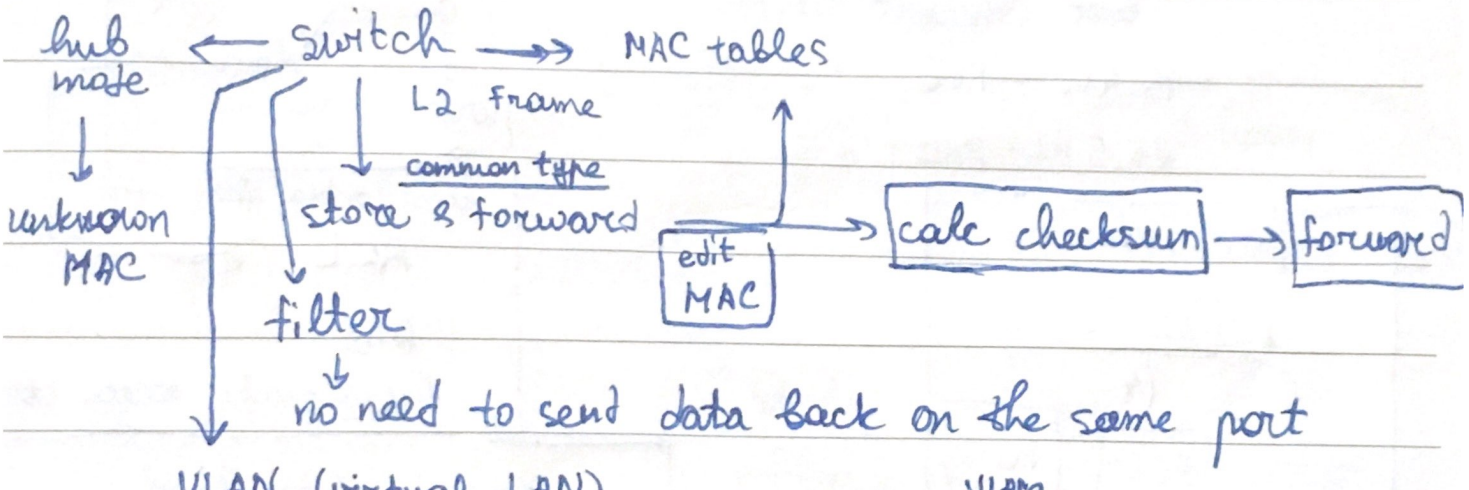


PDUs → protocol data unit (up)
SDUs → service data unit (down)
] OSI model

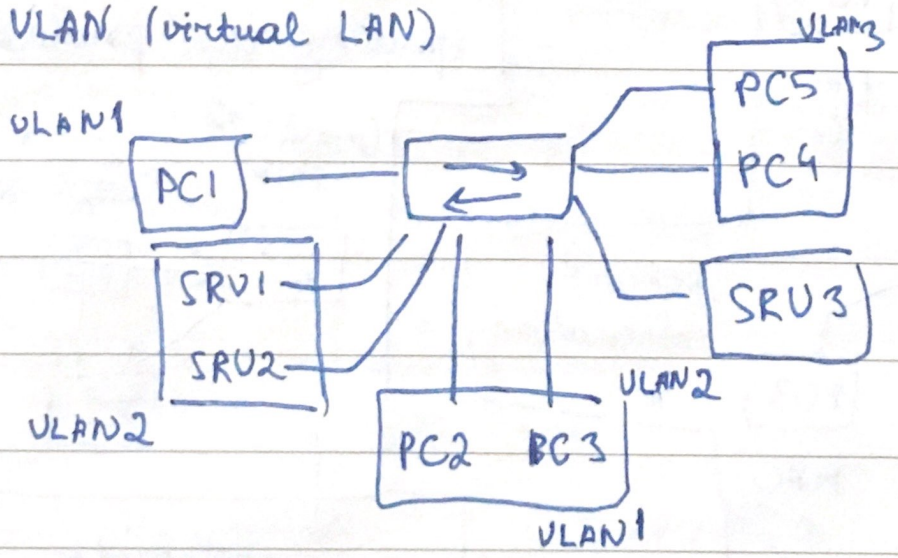
PDUs [
- Layer 5-7 = Data
- Layer 4 = Segment
- Layer 3 = Packet
- Layer 2 = Frame
- Layer 1 = Bit

Some People Fear Birthday

MTU - maximum transmission unit
(def: 1500 bytes)



VLAN (virtual LAN)



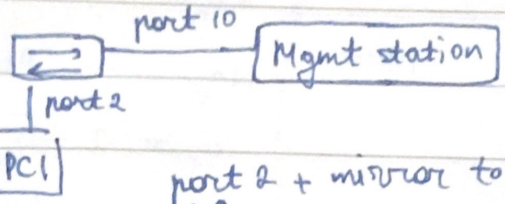
Отноо

802.1Q → VLAN tagging

tagged port → send data

untagged port → Native VLAN

port mirror →



port 2 + mirror to port 10

other

802.3af

PoE → power over ethernet

PoE+ 802.3at

send electrical current over Ethernet

15.4W - PoE

25.5W - PoE+

port forwarding

forward

port

basing on

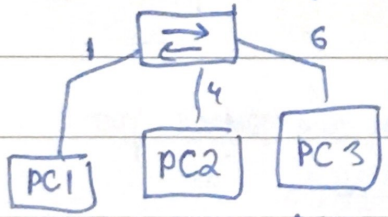
access control list

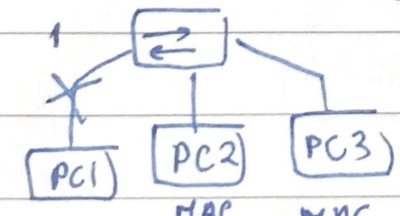
ACL

DAC

discretionary access ctrl

port security



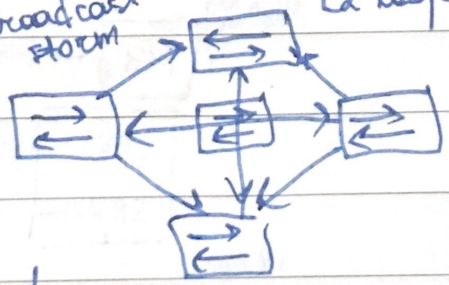


discard information

spanning tree protocol

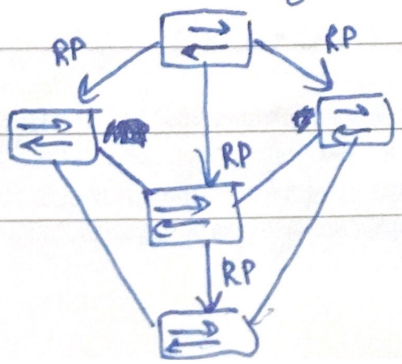
broadcast storm

L2 loop



root bridge

RP-root port



Огноо

distributed switching - SDN - software-defined networking

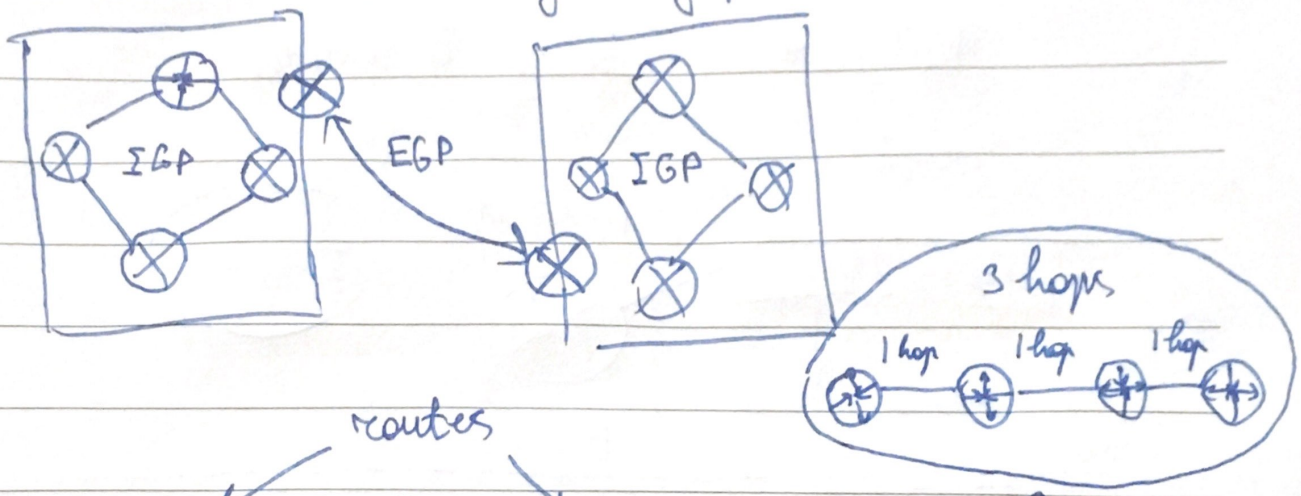
access layer → access

distribution layer → routing/switching

core layer → high speed

IGP - interior gateway protocols

EGP - exterior gateway protocols



routes

dynamic

static

× flexibility

hops

max 16 hops

IGP

RIP + RIPv2

not used

metric

dist. vector protocol

routing information protocol

sharing routing table 30s

Cisco-prop.

EIGRP

enhanced interior gateway routing protocol

convergence

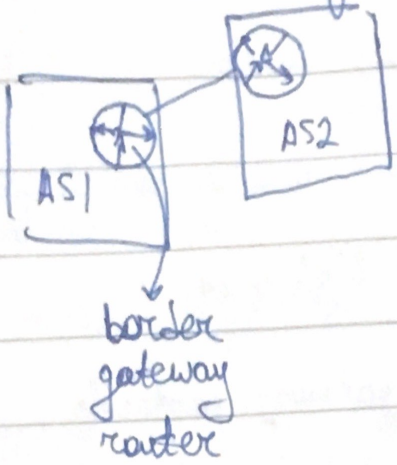
topology update

OSPF

open shortest path-first → Link-state routing protocol

bandwidth

BGP - border gateway protocol



NAT ⟶ ningling IP header

Private

